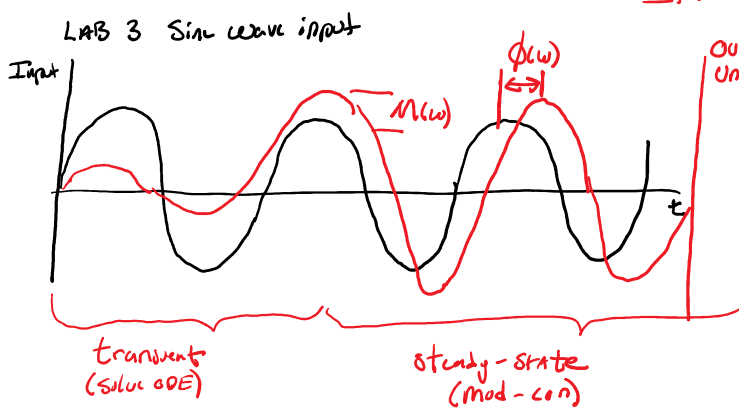
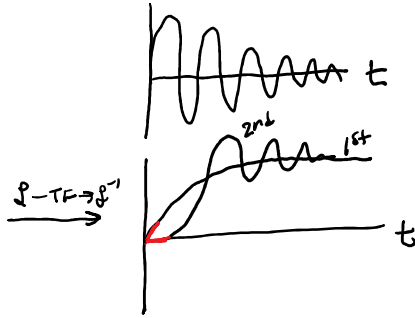
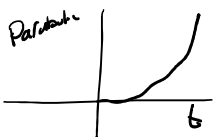
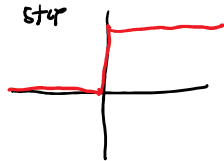
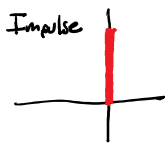




Inputs

Outputs

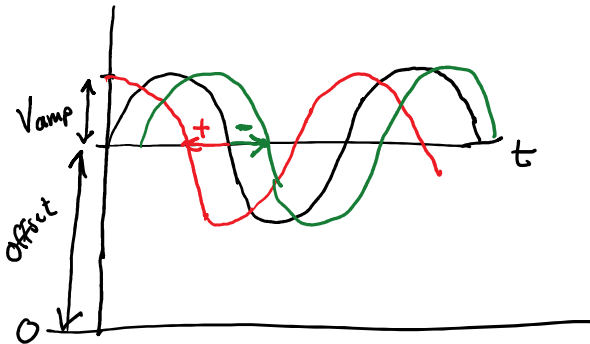


Input sine wave
out is sine wave
(in steady-state)

Two things change:

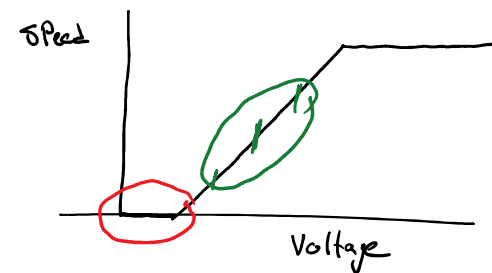
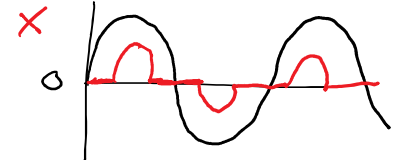
- Magnitude
 - Shift/delay/Phase/angle
- seconds radians.

$$V_{in} = V_{amp} \sin(\omega t) + \text{offset } t$$



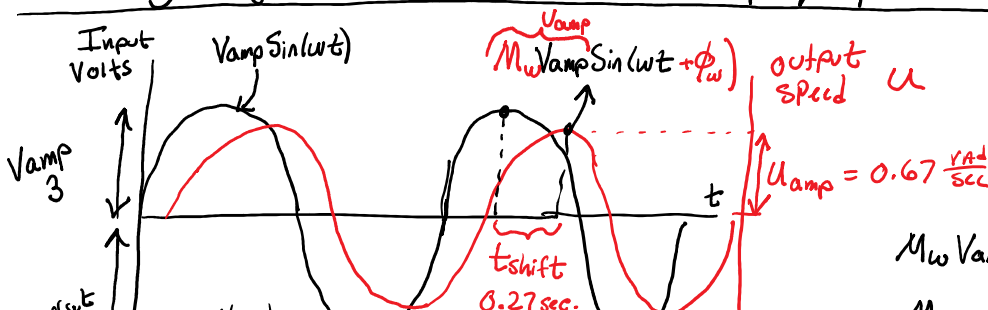
$\sin(\theta)$

$$\text{Angle} = \omega t$$

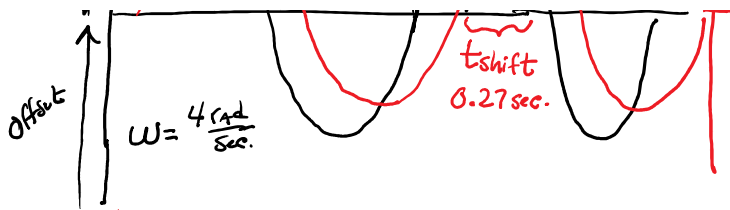


$V_{amp} \sin(\omega t + \pi/2) \rightarrow$ shifts Left $90^\circ \rightarrow$ happens before
 $V_{amp} \sin(\omega t - \pi/2) \rightarrow$ shifts Right $90^\circ \rightarrow$ happens After
 Negative if shift Right!

Extracting Magnitude and Phase from Input/output data (LAB 3)



$$M_w V_{amp} = u_{amp}$$



$$M_{\omega} V_{amp} = U_{amp}$$

$$M_{\omega} = \frac{U_{amp}}{V_{amp}}$$

$$\phi_{\omega} = -\omega t_{shift} = -\left(4 \frac{\text{rad}}{\text{sec}}\right)(0.27 \text{ sec}) = -1.1 \text{ rad}$$

↑ don't forget this!

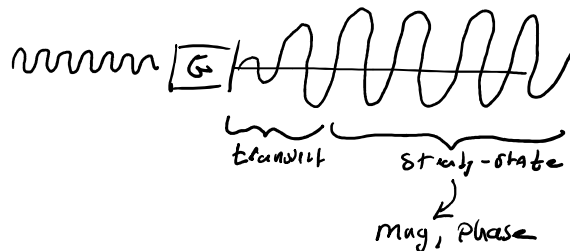
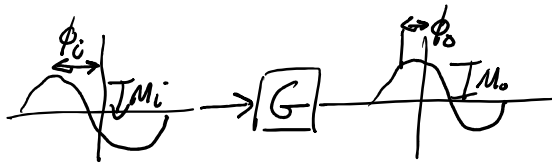
$$M_{\omega} = \frac{U_{amp}}{V_{amp}} = \frac{0.67 \text{ rad/sec}}{3 \text{ volts}} = 0.2236 \frac{\text{rad/sec}}{\text{volts}} = M_{abs}$$

Unit Conversion : $M_{dB} = 20 \log_{10}(M_{abs})$

$$M_{abs} = 10^{\left(\frac{M_{dB}}{20}\right)}$$

Predicting Frequency Response (Mod Con)

Frequency Response: Steady-state Response to input sin wave

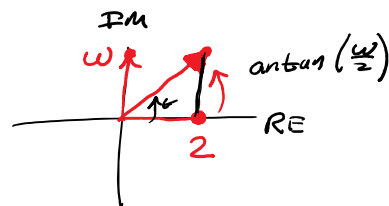
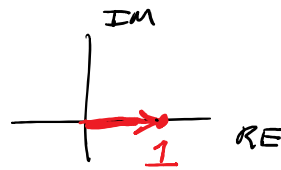


Frequency Response is defined as

$$G(s) \Big|_{s=j\omega} = G(j\omega)$$

Analytical Frequency Response:

Example $G = \frac{1}{s+2} \Rightarrow G(j\omega) = \frac{1}{j\omega+2}$



$$M(\omega) = \frac{1}{\sqrt{2^2 + \omega^2}} \xrightarrow{\omega=4} \frac{1}{\sqrt{2^2 + 4^2}} = 0.2236 \frac{\text{rad/sec}}{\text{volts}}$$

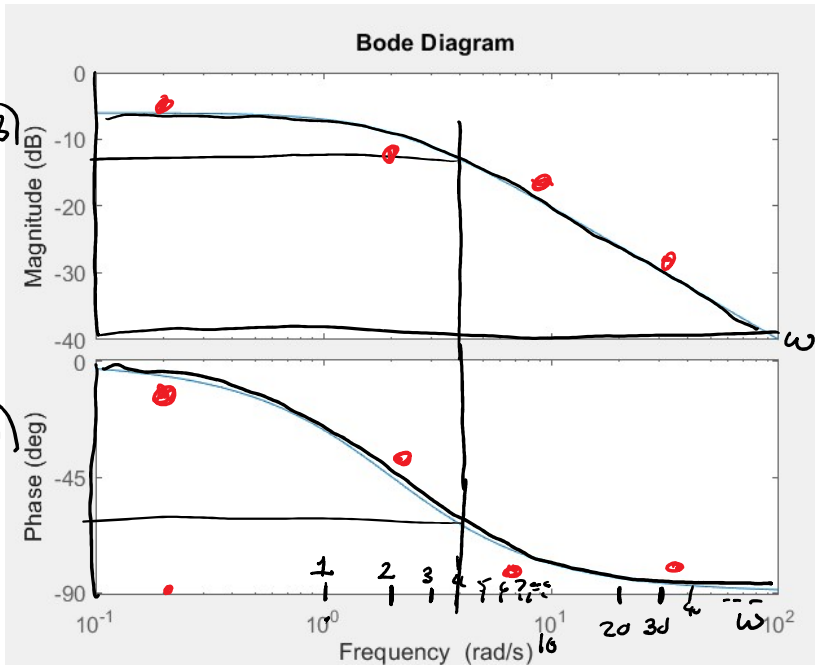
$$\phi(\omega) = \frac{-90^\circ}{\omega \tan(\frac{\omega}{2})} = 0 - \omega \tan(\frac{\omega}{2}) \xrightarrow{\omega=4} -\omega \tan(\frac{4}{2}) = -1.1 \text{ rad}$$

$$\frac{1}{2} \arctan\left(\frac{\omega}{2}\right)$$

Bode (G)

$M_{\omega}(\text{dB})$
mag

$\phi_{\omega}(\text{deg})$



$$M_{dB} = -13 \text{ dB}$$

$$M_{abs} = 10^{\left(\frac{-13}{20}\right)} = 0.223 \frac{\text{rad/s}}{\text{V}}$$

$$\phi = -65 \text{ deg} = -1.1 \text{ rad}$$